

HW_01

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a. State your assumptions and write the rules of the simulation in a manner similar to what I did in the notes. Here, we are assuming the following rules:

1. The data are normally distributed.
2. The data points are identically and independently distributed.
3. The true standard deviation is known for the Z test.
4. The true standard deviation is unknown for the t test.

Rules of the Simulation

1. Calculate the critical values for the t-test and Z-test using the user-defined confidence interval.
2. Generate normally distributed i.i.d. data using the user-defined mu and sd.
3. Calculate the standard error for both distributions based on assumptions.
4. Construct the confidence intervals for the t-statistic and Z-statistic.
5. Increase the individual counter for each distribution if the confidence interval contains the true mu.
6. Repeat steps 2 to 5 for R number of replicates.
7. Finally, divide the individual counters by the number of replicates to obtain true coverage probability.

b. Explain how these rules are translated into a computer program and how your function works.

Specify Parameters

```
p = target coverage probability of the CI
mu = true mean for the simulated data
sd = true standard deviation for the simulated data
N = number of sample
R = number of replicates
```

```
initialize t_cov = 0
initialize z_cov = 0
calculate t_crit = qt(1 - (1 - p) / 2, df = N - 1)
calculate z_crit = qnorm(1 - (1 - p) / 2)
```

Loop over replicates

```
{
  generate sample = Normal(mu, sd = sd, size = N)
  compute t_se = sd(sample) / sqrt(N) and z_se = sd / sqrt(N)
  construct the confidence interval:
  t_ci = [mean(sample) - t_crit * t_se, mean(sample) + t_crit * t_se]
  z_ci = [mean(sample) - z_crit * z_se, mean(sample) + z_crit * z_se]
```

```

    check true coverage:
    if (mu >= t_ci[1] and mu <= t_ci[2]) then t_cov += 1
    if (mu >= z_ci[1] and mu <= z_ci[2]) then z_cov += 1
    store the result
}
return t_cov/R and z_cov/R

```

```

CISim <- function(p=0.95, mu=0, sd=1, N=30, R=1000){

  # p: target coverage probability of the CI (e.g. p=0.95 for a 95% CI).
  # mu: population mean for the simulated data. Default value=zero.
  # sd: population standard deviation for the simulated data. Default value=one.
  # N: sample size for the simulated data. Default value=30.
  # R: number of simulation replicates. Default value=1000.

  t_cov = 0
  z_cov = 0

  # calculate critical values
  t_crit = qt(1 - (1 - p)/2, df = N - 1)
  z_crit = qnorm(1 - (1 - p)/2)

  for (i in 1:R){
    rv = rnorm(n=N, mean=mu, sd=sd)

    avg = mean(rv)
    # calculate standard errors
    t_se = sd(rv)/sqrt(N)
    z_se = sd / sqrt(N)

    # calculate confidence interval
    t_lower = avg - t_crit * t_se
    t_upper = avg + t_crit * t_se

    # calculate confidence interval
    z_lower = avg - z_crit * z_se
    z_upper = avg + z_crit * z_se

    # check the true coverage probability
    if (mu >= t_lower && mu <= t_upper) t_cov = t_cov + 1
    if (mu >= z_lower && mu <= z_upper) z_cov = z_cov + 1
  }
  prob = c(t_cov/R, z_cov/R)
  return(prob)
}

# get true coverage probability
prob = CISim()
{
  #print the output
  cat("method coverage\n")
}

```

```
cat(sprintf(" t %.3f\n", prob[1]))
cat(sprintf(" Z %.3f\n", prob[2]))
}
```

The function `CISim(p, mu, sd, N, R)` has been implemented as follows:

```
## method coverage
## t 0.949
## Z 0.942
```

c. Use your function to explore the performance of the t-based and Z-based confidence intervals as you vary different parameters. Does the coverage probability always approximate its target value? Write a short report To explore the true coverage probability for t-based and Z-based confidence intervals, we simulated the function `CISim()` using different combinations of parameters. For the first scenario, we increased the true mean sequentially in 0.5 increments to check the true coverage probability for both t and Z-statistics. The plot below shows that the true coverage remains close to 95% in most cases.

```
means = seq(0, 20, 0.5)

out = matrix(nrow = length(means), ncol=2) # matrix to store true coverage probability

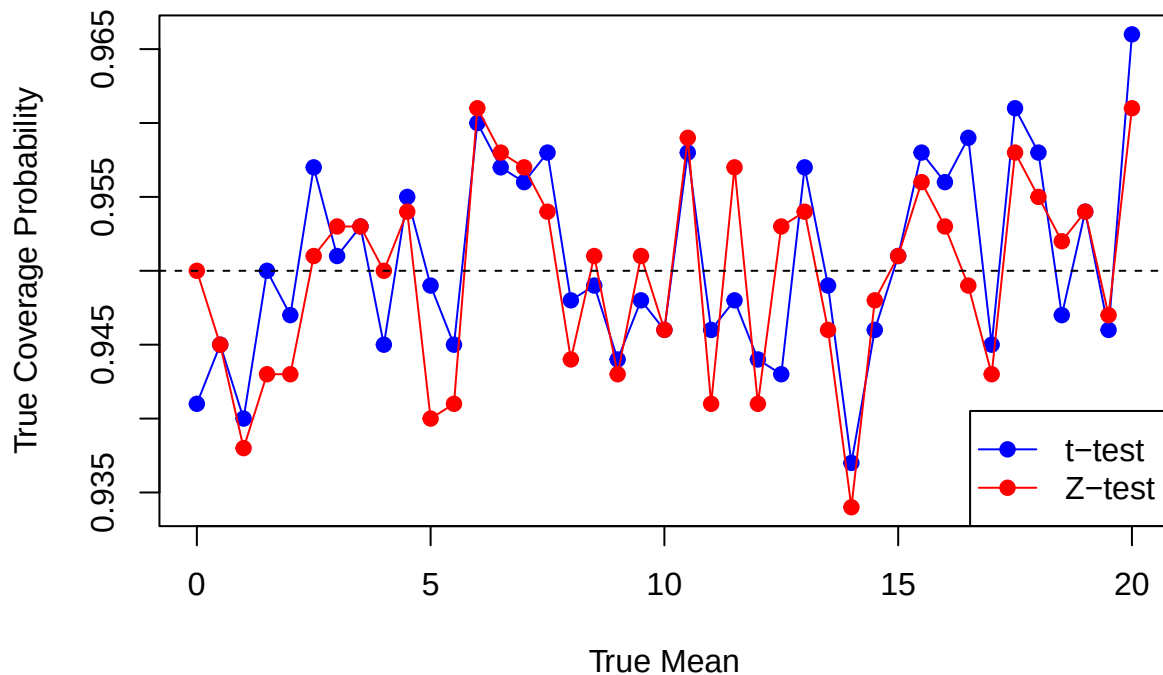
for (i in 1:length(means)){

  p = CISim (p=0.95, mu=means[i], sd=1, N=30, R=1000)
  out[i, 1] = p[1]
  out[i, 2] = p[2]
}

{
  # plot true coverage for t
  plot(means, out[,1], col="blue", pch=19, type = 'o',
       xlab = "True Mean", ylab = "True Coverage Probability",
       ylim = range(c(out[,1], out[,2])))

  # plot true coverage for Z
  lines(means, out[,2], col="red", pch=19, type = 'o')

  # 95% confidence interval
  abline(h = 0.95, lty = 2)
  legend("bottomright", legend = c("t-test", "Z-test"), col = c("blue", "red"), pch = 19, lty = 1)
}
```



Secondly, we kept all parameters constant at their default values except for the sample size, which was increased from 5 to 2000 in increments of 100. From the plot, we can see that for a small number of samples, the t-test and Z-test provide slightly different coverage probabilities. However, as the sample size increases, the true coverage for both tests becomes identical.

```

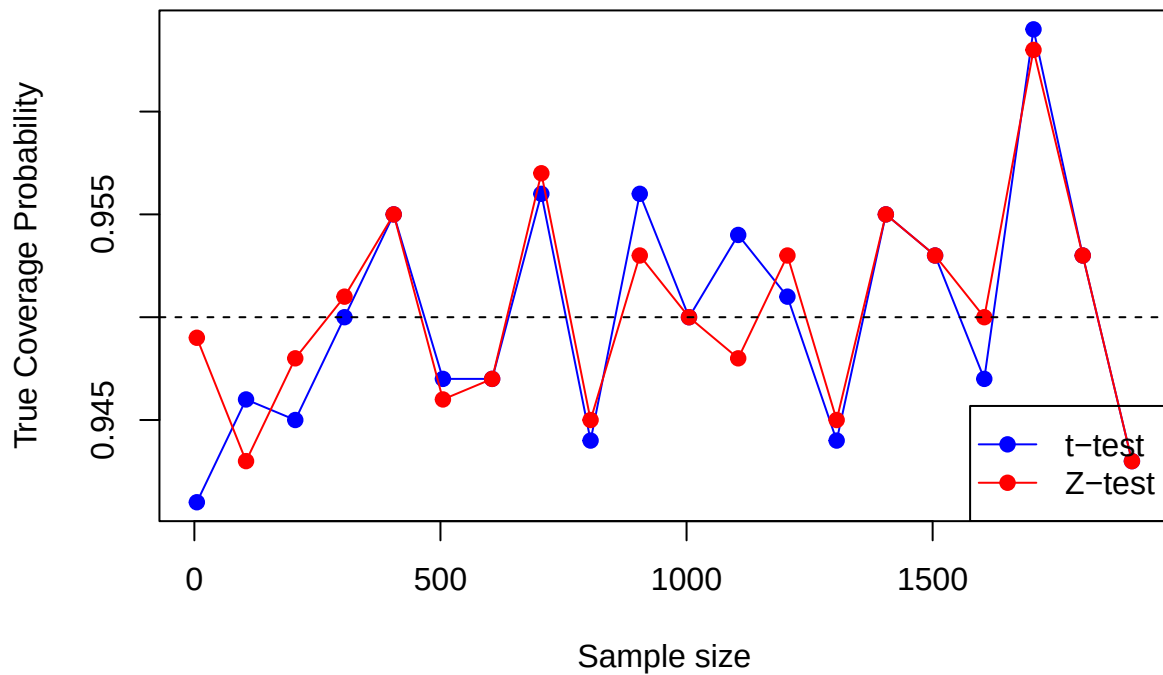
nsample = seq(5, 2000, 100)
out = matrix(nrow = length(nsample), ncol=2) # matrix to store true coverage probability
for (i in 1:length(nsample)){
  p = CISim (p=0.95, mu=0, sd=1, N=nsample[i], R=1000)
  out[i, 1] = p[1]
  out[i, 2] = p[2]
}

{
  # plot true coverage for t
  plot(nsample, out[,1], col="blue", pch=19, type = 'o',
       xlab = "Sample size", ylab = "True Coverage Probability",
       ylim = range(c(out[,1], out[,2])))

  # plot true coverage for Z
  lines(nsample, out[,2], col="red", pch=19, type = 'o')

  # 95% confidence interval
  abline(h = 0.95, lty = 2)
  legend("bottomright", legend = c("t-test", "Z-test"), col = c("blue", "red"), pch = 19, lty = 1)
}

```



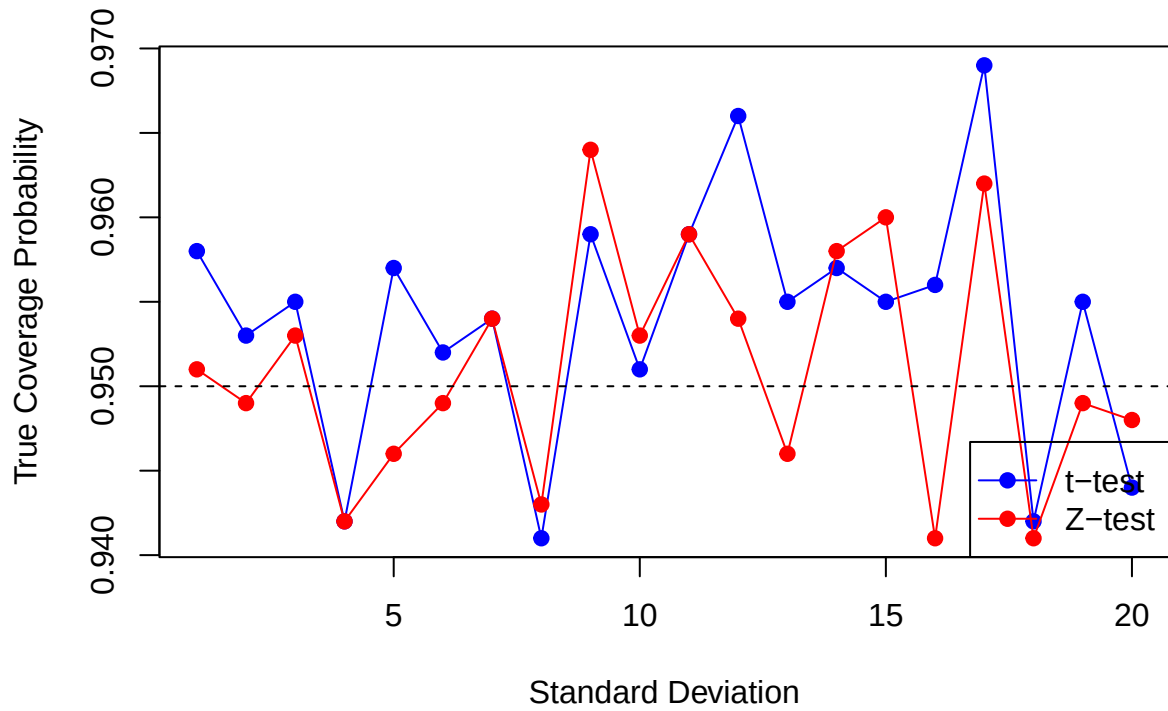
In my final simulation, we changed the standard deviation of the data from 1 to 20 in increments of 2. The output demonstrates a robust coverage of the parameter μ at the 95% confidence interval.

```
sd = seq(1, 20, 1)
out = matrix(nrow = length(sd), ncol=2) # matrix to store true coverage probability
for (i in 1:length(nsampl)) {
  p = CISim (p=0.95, mu=0, sd=sd[i], N=30, R=1000)
  out[i, 1] = p[1]
  out[i, 2] = p[2]
}

{
  # plot true coverage for t
  plot(sd, out[,1], col="blue", pch=19, type = 'o',
       xlab = "Standard Deviation", ylab = "True Coverage Probability",
       ylim = range(c(out[,1], out[,2])))

  # plot true coverage for Z
  lines(sd, out[,2], col="red", pch=19, type = 'o')

  # 95% confidence interval
  abline(h = 0.95, lty = 2)
  legend("bottomright", legend = c("t-test", "Z-test"), col = c("blue", "red"), pch = 19, lty = 1)
}
```



In conclusion, we simulated three different scenarios including different true means, standard deviations, and sample sizes to explore the performance of t-based and Z-based confidence intervals. Across all cases, the true mean was captured within the 95% confidence interval approximately 95% of the time for both t and Z statistics.